Integration of reciprocal functions

1 Write an equation for $f(x)$, given that the derivative of $f(x)$ is $\frac{1}{x}$.

2 Find the primitive function of the following:

a $\frac{-2}{x}$ b $\frac{6}{x}$ c $\frac{1}{7x}$ d $\frac{1}{x+5}$

3 Find the following indefinite integrals:

a $\int \frac{4}{x}\,dx$ b $\int \frac{-7}{x}\,dx$ c $\int \frac{3}{4x}\,dx$ d $\int \frac{1}{x-4}\,dx$

e $\int \frac{2}{x+3}\,dx$ f $\int \frac{2}{2x+3}\,dx$ g $\int \frac{-2}{x-4}\,dx$ h $\int \frac{1}{2x+3}\,dx$

i $\int \frac{1}{5-x}\,dx$ j $\int \frac{2}{3x+5}\,dx$ k $\int \frac{2x}{x^2-5}\,dx$ l $\int \left(\frac{8}{x}+5\right)\,dx$

4 Find the exact value of the following definite integrals:

a $\int_3^9 \frac{3}{x}\,dx$ b $\int_4^6 \frac{1}{x-2}\,dx$ c $\int_3^5 \frac{2}{x+2}\,dx$ d $\int_{\frac{8}{3}}^{\frac{17}{3}} \frac{4}{3x-2}\,dx$

e $\int_1^3 \frac{5}{x+1}\,dx$ f $\int_3^5 \frac{x^2+1}{x}\,dx$ g $\int_0^3 \frac{1}{5+4x}\,dx$ h $\int_2^6 \left(2x+\frac{1}{x+2}\right)\,dx$

5 Determine the integral of the following functions:

a $f(x) = \frac{2x^2+x-3}{x}$, for $x > 0$. b $f(x) = \frac{5x^3+2x^2}{x^3}$, for $x > 0$.

6 Consider the function $f(x) = x \ln x$.

a Differentiate $f(x) = x \ln x$. b Hence, find $\int \ln x\,dx$.

Reverse chain rule

7 Consider the function $f(x) = \ln(x^2 - 4)$.

a Use the chain rule to differentiate $f(x)$.

b Hence, find $\int \frac{6x}{x^2-4}\,dx$.

8 Consider the function $f(x) = \ln(2x^2 + 4x)$.

a Use the chain rule to differentiate $f(x)$.

b Hence, find $\int \frac{x+1}{2x^2+4x}\,dx$.

9 Consider the function $f(x) = \ln(3x^3 + 5x^2)$.

a Use the chain rule to differentiate $f(x)$.

b Hence, find $\int \frac{18x^2+20x}{3x^3+5x^2}\,dx$.

10 Find the integral of the following functions:

a $\int \frac{4}{4x-3}\,dx$

b $\int \frac{15}{3x-2}\,dx$

c $\int \frac{-8}{2x+1}\,dx$

d $\int \frac{4x}{x^2+1}\,dx$

e $\int \frac{3x^2}{x^3+1}\,dx$

f $\int \frac{8x}{2x^2-3}\,dx$

g $\int \frac{2x^2}{2x^3+7}\,dx$

h $\int \frac{6x^2+3}{2x^3+3x}\,dx$

Area under the curve

11 Consider the function $y = \frac{2}{x}$.

a Sketch a graph of the function for $x > 0$.

b Find the exact area bounded by the curve, the x-axis, and the lines $x = 2$ and $x = 3$.

12 Consider the function $y = \frac{1}{x-2}$.

a Sketch a graph of the function for $x > 0$.

b Find the exact area bounded by the curve, the x-axis, and the lines $x = 3$ and $x = 5$.

13 Consider the function $y = \frac{1}{x+3}$.

a Sketch a graph of the function $y > 0$.

b Find the exact area bounded by the curve, the x-axis, and the lines $x = 2$ and $x = 4$.

14 Consider the function $y = \dfrac{1}{2x+5}$.

- **a** Sketch a graph of the function $y > 0$.
- **b** Find the exact area enclosed by the curve, the x-axis, and the lines $x = 3$ and $x = 4$.

15 Consider the function $y = \dfrac{1}{4-3x}$.

- **a** Sketch a graph of the function.
- **b** Find the exact area enclosed by the curve, the x-axis, and the lines $x = 3$ and $x = 5$.

16 Consider the function $y = \dfrac{2}{3x+5}$.

- **a** Sketch a graph of the function $x > 0$.
- **b** Find the exact area enclosed by the curve, the coordinate axes and the line $x = 4$.

17 Consider the function $y = x + \dfrac{1}{x}$.

- **a** State the domain of the function.
- **b** State the value the function approaches as $x \to \infty$.
- **c** Calculate the exact area enclosed by the function, the x-axis, and the lines $x = 2$ and $x = 12$.

18 Find the exact area enclosed by the function $f\left(x\right) = x + \dfrac{5}{x+5}$, the x-axis and the lines $x = 0$ and $x = 2$.

19 Consider the function $f\left(x\right) = \dfrac{6x^5 - 4x^2 + x}{2x^2}$, for $x > 0$.

- **a** Simplify and hence determine $\int f\left(x\right)\, dx$.
- **b** Find the exact area bounded by $f\left(x\right)$, the x-axis and the lines $x = 1$ and $x = 2$.

20 Consider the functions $y = x$ and $y = \dfrac{3}{x}$, for $x > 0$.

- **a** Sketch the graphs of these functions on the same coordinate axes.
- **b** At what x-value do the two functions intersect?
- **c** Find the area between the two functions, the x-axis and the line $x = 3$.

21 Consider the functions $y = -x^2$ and $y = \frac{\text{[illegible]}}{4-x}$ for $0 \leq x \leq 3$.

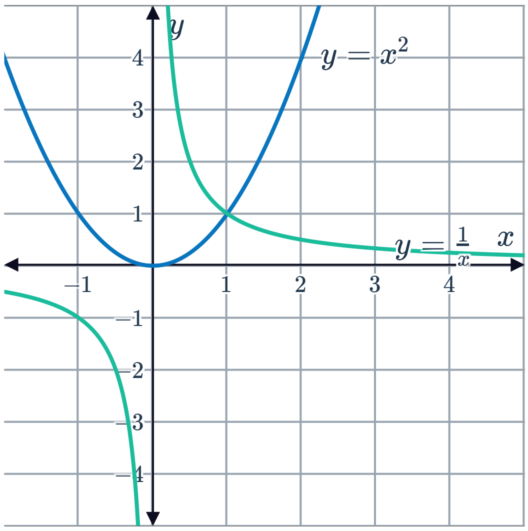
- **a** Sketch the graphs of these functions on the same coordinate axes.
- **b** Calculate the exact area bound by the curves between $x = 1$ and $x = 3$.

22 Consider the functions $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x}$.

- **a** Sketch the graphs of the two functions on the same coordinate axes.
- **b** At what x-value do the two functions intersect?
- **c** Find the area of the region bounded by the two curves, the y-axis, and the line $y = 2$.

23 Consider the graph of the functions $y = x^2$ and $y = \frac{1}{x}$:
Solve for the exact value of k such that:

$$\int_1^k \frac{1}{x}\, dx = \int_0^1 x^2\, dx$$



24 Consider the function $f(x) = \ln\left(x^2 + 1\right)$.

- **a** Differentiate $f(x)$.
- **b** Hence, find $\int \frac{x}{x^2 + 1}\, dx$.
- **c** Find the area bounded by $g(x) = \frac{x}{(x^2 + 1)}$, the x-axis and the lines $x = 2$ and $x = 3$.

25 Consider the function $f(x) = \ln(\sin(x) + 1)$.

- **a** Differentiate $f(x)$.
- **b** Hence, find $\int \frac{\cos x}{\sin x + 1}\, dx$.
- **c** Find the area bounded by $g(x) = \frac{\cos x}{\sin(x) + 1}$, the x-axis and the lines $x = 0$ and $x = \frac{\pi}{6}$.